

Faculty of Computing

Year 2 Semester 1 (2026)

IT2011 - Artificial Intelligence and Machine Learning

Final Evaluation - Implementation

This stage focuses on Model Design, Implementation, and Evaluation Preparation using your assigned dataset, and will be assessed through a viva presentation.

Progress Review II – Viva Details

- Mode: In-class (during scheduled lab/workshop sessions)
- Duration: 20 minutes per group
- Format: Viva-style explanation and demonstration of models

Individual Requirements (45 marks per student)

Each student must:

- Design and implement at least one machine learning model, such as:
 - Classification (e.g., SVM, Decision Tree, Logistic Regression)
 - Clustering (e.g., K-Means, DBSCAN)
 - Dimensionality Reduction (e.g., PCA)
 - Deep Learning (e.g., MLP, CNN using Keras/TensorFlow)
- Explain your model by covering:

Model suitability to the assigned dataset and chosen problem

Implementation details (libraries, functions, hyperparameters)

Parameter tuning methods (e.g., GridSearchCV, manual tuning)

The number and variety of models trained by tuning parameters

The evaluation metric(s) for each variety of model trained, why they are appropriate, the validation method used, a comparison across varieties with a conclusion, and any limitations or improvements suggested

Note: Each student must evaluate at least one model, assessing performance across different varieties of the model with different pre-processing and hyperparameter tune-ups. If a student fails to present or demonstrate individual contribution, they will receive zero for this review.

- Be ready to demonstrate your section of the code and show output

Important: Students who do not implement and explain at least one model will receive zero for the individual section.

Group Requirements (10 marks for the group)

Whole group must:

1. Compare the optimum results gained from 6 different individual models trained by each member
2. Discuss challenges, performance, or expected behavior